Delft University of Technology

Computational Transport Phenomena

CH3421

Assignment 4

Date: 04/03/2026

Q1 - A fluid enters a pipe of radius 0.06 meters at a constant velocity of 0.1 m/s. The fluid has a density of 1.2 kg/m^3, a thermal conductivity of 0.02 W/mK, a specific heat of 1000 J/ $(kg \cdot K)$, and a viscosity of $1.8 \cdot 10^{-5}$ $kg/(m \cdot s)$. The first 5.76 meters of the pipe are isothermal, held at 300 K. The remaining 2.88 meters of the pipe have a constant heat flux of 37.5W/m^2 added at the wall.

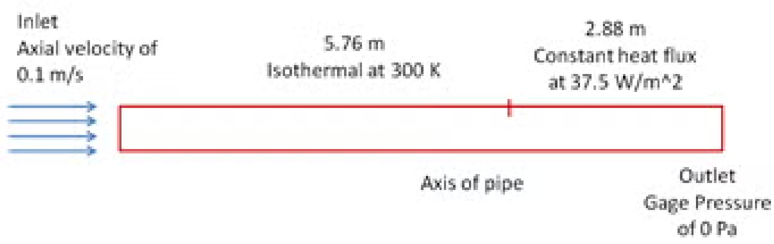

Figure 1: Sketch of the laminar flow in a pipe.

Using Ansys Fluent, simulate the above flow. Calculate and plot the velocity, temperature, pressure and Nusselt number variation in the pipe.

Q2 - Now refine the mesh and repeat the calculation.